

SSH Key-Based Authentication and Disable Password Login

Question 1: Install OpenSSH server and verify sshd service.

1. Install OpenSSH Server:

Ubuntu/Debian: `sudo apt update && sudo apt install openssh-server -y`

RHEL/CentOS/Amazon Linux: `sudo yum install openssh-server -y`

2. Start Service: `sudo systemctl start sshd`

3. Enable Service: `sudo systemctl enable sshd`

4. Check Status: `sudo systemctl status sshd`

5. Test: `ssh localhost`

Question 2: Generate SSH key pair and configure authorized_keys.

1. Generate Key: `ssh-keygen -t rsa -b 4096`

2. View Key: `cat ~/.ssh/id_rsa.pub`

3. Add Key: `cat ~/.ssh/id_rsa.pub >> ~/.ssh/authorized_keys`

4. Set Permissions:

`chmod 700 ~/.ssh`

`chmod 600 ~/.ssh/authorized_keys`

5. Test: `ssh localhost`

Question 3: Disable password authentication.

1. Edit file: `sudo vi /etc/ssh/sshd_config`

2. Change PasswordAuthentication yes to PasswordAuthentication no

3. Ensure PubkeyAuthentication yes

4. Restart SSH: `sudo systemctl restart sshd`

5. Verify: `sudo systemctl status sshd`

6. Test key login: `ssh localhost`

7. Test password login: `ssh -o PreferredAuthentications=password localhost`

Expected: Permission denied (publickey)

Question 4: Documentation

Document all commands used, configuration changes, and screenshots of installation, sshd status, key generation, authorized_keys setup, sshd_config changes, successful SSH login, and failed password login.

Conclusion

OpenSSH was installed, SSH keys were configured, password authentication was disabled, and login was verified using SSH key-based authentication only.